⚛ MISCELLANEOUS — FULL MIND MAP

विविध · Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Physics Branches, Scientific Institutions, LASER, Inventions & ATM

- Quantum Mechanics physics ki branch hai jo atoms aur subatomic particles jaise bahut chhote particles ki motion aur behavior se related hai. Geodesy Earth ke teen basic properties ko accurate roop se measure karne aur samajhne ka science hai: iska geometric shape (dimensions), space me iska orientation; iska gravity field, saath hi time ke saath in properties me parivartan (elevation/depression). Isliye, ek se zyada option sahi hain. *[Q1–Q2]*
- Cybernetics mechanical, physical, biological, cognitive aur social systems ke study se relevant hai. Norbert Wiener ne 1948 me Cybernetics ko "animal aur machine me control aur communication ke scientific study" ke roop me define kiya tha. Horology time measure karne ki art ya science hai. Isme clocks, sundial, timer aur atomic clocks etc. ka study shamil hai. *[Q3–Q4]*
- Tribology mechanical engineering aur Material Science ki ek branch hai. Isme Friction, wear aur lubrication ke principles ka study aur application shamil hai. Plasma ko aksar 'matter ki fourth state' kaha jata hai. Isme charged particles (ions aur free electrons) ki significant amount hoti hai. Isliye statement (i) true hai. *[Q5–Q6]*
- 'White Dwarf' (white dwarf) ek stellar core ka remnant hai jo mainly electron-degenerate matter se bana hota hai. Ye un stars ki last evolutionary state ko represent karta hai jinka mass neutron star banane ke liye enough nahi hai. naturally, iska study Astronomy me kiya jata hai. Statement (B) incorrect hai. Rayleigh scattering ke according, chhoti wavelength zyada scatter hoti hai. Violet light ki wavelength blue light ke comparison me chhoti hoti hai, isliye violet waves blue light waves se zyada scatter hoti hain. Sky violet ke bajaay blue mainly isliye dikhai deta hai kyonki human eye blue light ke prati zyada sensitive hoti hai aur Sun violet ke comparison me zyada blue light emit karta hai. *[Q7–Q8]*
- India ki National Physical Laboratory New Delhi me located hai. Ye India ki measurement standards laboratory hai, jo country me SI units ke standards ko banaye rakhti hai aur weights and measures ke National standards ko calibrate karti hai. National Chemical Laboratory (CSIR-NCL), jiski establishment 1950 me hui thi, Council of Scientific and Industrial Research (CSIR) ki ek constituent laboratory hai. Ye Pune, Maharashtra me located hai. *[Q9–Q10]*
- Tata Institute of Fundamental Research (TIFR) mathematics aur science me advanced research ke liye ek major institute hai. Iski establishment 1945 me Sir Dorabji Tata Trust ke sahayog se Dr. Homi Bhabha dwara ki gayi thi; ye Mumbai me located hai. LASER ka full form "Light amplification by stimulated emission of radiation." hai. Ye ek aisa device hai jo electromagnetic radiation ke stimulated emission ke basis par optical amplification (optical amplification) ki process ke through light emit karta hai. *[Q11–Q12]*
- Choonki LASER ka full form hi "Stimulated radiation ke emission dwara light pravardhana" hai, lejara naturally Stimulated radiation ke production ke liye design kiya gaya ek device hai. Ruby LASER ek solid-state LASER hai jo synthetic ruby kristal ka use karta hai. Pehla working LASER thiodor echa. Maiman dwara 1960 me banaya gaya ek ruby LASER tha. *[Q13–Q14]*
- Ek Japanese author aur freelance journalist Takashi Hirose ne 2010 me 'nyookliyara riektar taaima bama' book likhee thi. Revolver ka invention saimual kolta ne kiya tha (A-3). Dynamite ka invention Alfred Nobel ne kiya tha (B-1). *[Q15–Q16]*
- Usually, phlaishalaaita/torcha me standard shushk sela (leklaanche sela) ka use kiya jata hai. Ina koshikaaon me, enoda (nakaaraatmaka tarminala) ek jastaa (jinka) kaina se bana hota hai, jabki kaithod (sakaaraatmaka tarminala) ek kaarban (grephaaita) roda hota hai. E.tee.ema. (A.T.M.) ka full form 'Automated Teller Machine' hai. Ye ek vishesh computer hai jo bainka khaate ke phanda ko prabandhit karna suvidhaajanak banata hai, jiska use usually paise nikaalane ke liye kiya jata hai. *[Q17–Q18]*
- Neshanal phaainenshiyal svicha (NFS) India me share automatic telara machineon (ATM) ka sabse large network hai. Ye Bhartiya National bhugataan nigama (NPCI) dwara chalaay jata hai, jisne 2009 me IDRBT se apanaa sanchaalan sanbhaala tha. Isliye, NPCI country ke sabhi eteeem ko connect karta hai. *[Q19]*

⚠ EXAM TRAP — Q2 & Q14 — Wording Matters

▸ Q2 me Geodesy sirf Earth ke dimensions tak limited nahi hai; shape, gravity field aur orientation bhi broader definition ka part hain. Source therefore multiple aspects ko accept karta hai.
▸ Q14: first working ruby laser (Theodore Maiman, 1960) deep-red coherent light (~694 nm) produce karta hai — blue-light statement incorrect hai.

2. Nanotechnology: Nanoparticles, CNTs, Quantum Structures & Nano Devices

- Nanotechnology nanoskel par organized vijnyaana, engineering aur technology hai, jo lagbhag 1 se 100 nanomeetar hai. Ye nuclear aur aanavik paimaane par matter ke herapher ki allow karta hai, jo devices ke laghukaran me important bhoomika nibhaata hai. ek nanoparticle ke roop me classified hone ke liye ek nano-objekta ke specific aayamon me se ek 1-100 nm (jo ki 10 -9 se 10 -7 meetar hai) ki seemaa me hota hai. Isliye, 10 -7 meetar se kam dimensions wala particle ek nanoparticle hai. *[Q20–Q21]*
- Nanoteknolojee aur nanoparticle ki featureen nanoskel par built matter par laagoo hoti hain, jise world level par 1 se 100 nanomeetar (nm) ke beech define kiya gaya hai. Statement 1 incorrect hai kyonki nanoparticle praakritik roop se jvaalaamukheey raakha, oceanee phuhaara, mahina reta aur jaivik processon (jaise vaayarasa) me paae jate hain. Statement 2 sahi hai. Zinc oxide aur taaiteniyam daaioksaaid jaise dhaatu oksaaid ka widely sanaskreen aur saundary prasaadhanon me use kiya jata hai. *[Q22–Q23]*
- 'nano plaga' (Nano plug) ek highly chhote, lagbhag avisible shravan device (sunane ki machine) ko refer karta hai. Isme microscopic component aur ek nano-Battery shamil hoti hai, jo kaana nahara ke andara depression me phita baithatee hai. 2012 me, scientists ne ek vaikalpik roop se phanse sone (gold) ke nanoparticleon ke basis par pehla 'nanoiyara' (nanoear) developed kiya. Ye microscopic device -60 dB jitane kam Sound/kanpana staron ka detect lagaa sakta hai. *[Q24–Q25]*
- Tokyo worldvidyalay ke prophesar noriyo taaniguchee ne pehli baar 1974 me apne shodha patra me "nanoteknolojee" word ka istemaal kiya tha. Kaarban nanotyoob (CNTs) ne lakshit davaa vitaran vaahak ke roop me utkrisht capacity dikhai hai (statement 1) aur jaiva chemical sensar me effective dhanga se use kiye jate hain (statement 3). Adhyayanon se ye bhi detect chalaa hai ki specific enjaaim kuch type ke CNTs ko baayodigred (nashta) kar sakte hain (statement 4). Haalaanki, kachche kaarban nanotyoob ganbhir rakta ke thakke ka kaaran bana sakate hain aur vartamaan vyavasaayik level par kritrim rakta keshikaaon ke roop me directly use ke liye rakta-sangata nahi maane jate hain, jisse is sandarbh me statement 2 incorrect hai. *[Q26–Q27]*
- Nanomedisin aise nanoparticleon ko janma deti hai jinake through behatar lakshit specific davaa vitaran (jaise, directly kainsar koshikaaon tak) sanbhav hai. Isliye statement 1 sahi hai. Nanotechnology shareer me ksharan (degradation) se bachate hue jeena ko koshikaaon me safe roop se pahunchaane me bhi highly effective hai, jo jeena therepee ko aage badhane me large yogadaan deta hai. Isliye statement 2 bhi sahi hai. Aaeeseeeaara-sarkota (ICAR-CIRCOT) Mumbai dwara established India ka pehla nano-selyuloja sandevice, high moolya wale nano-selyuloja taiyar karne ke liye kachche maala ke roop me Cotton (cotton) lintars aur Cotton ke kachare ka use karne ke liye specific roop se design kiya gaya hai. *[Q28–Q29]*
- Nanoskel par, physical property naatakeey roop se badala jate hain. Nanomaiteriyals ka ek major, heavy study kiya jane wala property Friction hai (visheshakara nanoTribology me), kyonki high surface-se-aayatana anupaat maulik roop se badala deta hai ki material Ek-doosare ke khilaaph kaise slaaid karti hai. Nanoskel tak seemit aayamon ke basis par nano technology classification me: 1 dimensions pratibandhit (2D free) = Quantum well (patalee philmen/paraten). 2 dimensions pratibandhit (1D free) = Quantum wire (nanotyooba). 3 dimensions pratibandhit (0D free) = Quantum dot. *[Q30–Q31]*
- Quantum dot kritrim roop se built A semiconductor nanostructure hai (usually kuch nanomeetar shape me) jo chaalan band electrons ki speed ko dridhata se seemit karta hai, jiske parinaamasvaroop adviteeya, aakaara-nirbhara optical aur electronic property hote hain. Pehli generation: nishkriy nanostrakchar jaise eyarosola, polimara. (A) Doosri generation: active nanostrakchar jaise lakshit davaaen. (D) *[Q32–Q33]*
- Nano hamingabard ek small, rimota-radioactivea nano air vaahan (NAV) hai jise hamingabard (Ek type ki chhoti chiriya) ke same dikhane aur udane ke liye design kiya gaya hai. Ise nigaraanee aur jaasoosee missionon ke liye eyarovironament dwara America me develop kiya gaya tha. *[Q34]*

⚠ EXAM TRAP — Q27 & Q30 — Nanotechnology Source Framing

▸ Carbon nanotubes ke biomedical uses material functionalization aur safety par depend karte hain; exam statement ko exact wording ke context me judge karein.
▸ Q30 ka “most important property” wording source-specific hai. Nanomaterials ki properties size, surface-area/volume ratio, quantum effects aur application ke according vary karti hain.

3. Energy Projects, Science Congress, Awards & S&T Organizations

- Jilaa hiting system ke liye bhoomigat bhootaapeey garmee (geothermal heat) ka dohana karne ka pehla successful modern prayas 1890 me Boise, Idaho (USA) (yooesae) shahar me kiya gaya tha, jahaan aasapaas ki imaaraton ko garma karne ke liye garma water ke paaipa network develop kiya gaya tha. Highly vivaadaaspad enaron bijalee project (originally Dabhol paavar kanpanee dwara prabandhita) India ke Maharashtra ke ratnaagiree jile me Dabhol me located hai. *[Q35–Q36]*
- Amarkantak power station Madhya Pradesh ke anoopapur jile ke chachaaee me located hai. Ye ma.pra. Paavar janareshan kanpanee limited (MPPGCL) ke koyala based bijalee steshanon me se ek hai. 88vaan Bhartiya science kaangres (ISC) sammelan 3 se 7 janavaree 2001 ke beech New Delhi me organized kiya gaya tha. Is sammelan ka main vishay 'khaadya, poshan aur paryavaran security' tha. Nota: janavaree 2023 me, 108veen ISC 'mahilaa sapowerkaran ke saath satata development ke liye science aur technology' ke vishay ke saath R.T.M. Naagapur worldvidyalay me organized ki gayi thi. *[Q37–Q38]*
- 26 September, 2006 ko tatkaaleen pradhaan mantree do. Manamohan sinha ne science bhavana, New Delhi me central chamada research institute (CLRI), Chennai ko graameen development ke liye S&T navaachaaron ka CSIR award provide kiya tha. Science aur engineering ke field me padma vibhooshana-2023 America ke shree esa.aara. Shreenivaas vardhan ko provide kiya gaya. Wo ek Bhartiya-American ganitajny hain. Unhe praayikata principle (probability theory) aur specially bade vichalanon (large deviations) ka integrated principle banane me unke maulik yogadaan ke liye jana jata hai. 2007 me, wo ebela award (Abel Prize) jeetane wale pehle eshiyaee bane the. *[Q39–Q40]*

- DST – science aur technology department (Department of Science and Technology), India government. CSIR – scientist aur industrial research parishada. Neshanal inoveshan phaaundeshana-indiya (NIF) India government ke science aur technology department ka ek svaayatt nikaay hai. Isliye statement 1 sahi hai. Isane maarch 2000 me jameenee level (grassroots) ke technical navaachaaron aur utkrisht conventional jnyaan ko majaboot karne ke liye India ki National pahala ke roop me kaam karna start kiya, na ki specially videshee instituteon ke saath advanced research ke liye. Isliye, statement 2 incorrect hai. *[Q41–Q42]*
- Shevaron (Chevron) ek American bahuNational energy nigama hai jo tela (Oil), praakritik gas etc. Me lagaa hua hai (A-2). AT&T ek American bahuNational telecom nigama (Telephone, Internet) hai (B-3). *[Q43]*

4. Polar/Ocean Research, Weather & Earth-Science Concepts

- Dakshin Gangotri Antarctica me established India ka pehla sthayee research hub tha. Ise 1984 me banaya gaya tha aur baad me 1990 me barpha me dabane ki wajah se ise sevaamukt (decommissioned) kar diya gaya tha. Jaisaa ki pehle bataay gaya hai, Antarctica me pehla sthayee Bhartiya research basis 'Dakshin Gangotri' tha. *[Q44–Q45]*
- Antarctica me India ka teesra aur sabse new sthayee research hub 'Bharti' (Bharti - 2012 me chaaloo) hai. Dakshin Gangotri pehla (1984) tha; Maitri doosra (1989) tha. Jaisaa ki oopara bataay gaya hai, Antarctica me India dwara established teesare research hub ka naam 'Bharti' hai. *[Q46–Q47]*
- IndARC (indaaarka) India ki pehli maltee-sensara water ke neeche (underwater) located observatory hai. Ise 23 July 2014 ko Northee dhruva aur norve ke beech lagbhag aadhe raaste me Arctic ke kongsaphajordan me successfultaaEastka tainaat kiya gaya tha. Pradhaanamantree narendr modee ne August 10, 2020 ko video konphrensing ke through Chennai-andamaana aur nikobaar dveepa group (CANI) submarine optical phaaibar kebala (OFC) ka shubhaaranbh kiya aur raashtr ko dedicated kiya. Ye project dveepon ko high speed (high speed) Internet kanektivitee provide karti hai. *[Q48–Q49]*
- 18 June 2023 ko, oshanaget (OceanGate) dwara operated ek deeper ocean me jane wali submarine, Titan, Northee atalaantik mahaasaagar me taaitainik ke malabe ko dekhane ke ek abhiyan ke dauran phata (imploded) gayi thi. 'polara code' (Polar Code) international oceanee organisation (IMO) ke dhruveey water me sanchaalan karne wale jahaajon ke liye international code ko refer karta hai. Ye 1 janavaree 2017 ko laagoo hua aur donon dhruvon ke aasapaas ke durgam water me chalne wale jahaajon ke designa, development aur paryavaran sanrakshan ko kavara karta hai. *[Q50–Q51]*
- Janavaree 2021 me naineetaal jile ke Mukteshwar me Uttarakhand ke pehle dopalar vedara radar (DWR) ka udghaatan kiya gaya. Ye toophaan aur heavy baarish jaisee ganbhir weather sthitiyon ki bhavishyavaanee karne ke liye air me varsha ki teevrata ko padha sakta hai. Samalavan line (Isohaline) maanachitr par wo line hai jo ocean me same lavanata (salinity) wale binduon ko connect karti hai, na ki barphabaaree ko. Isliye ye incorrect matched hai. Samapressure line same vaayumandaleey Pressure wale binduon ko connect karti hai, samochch line same elevation ko aur samagaharee line same water ke neeche ki depression wale binduon ko connect karti hai. *[Q52–Q53]*
- Pycnocline ocean ya water ke other pinda me ek layer hai jahaan depression ke saath water ka ghanatv fastee se badalt hai. Isliye, ye oceanee water ghanatv ke saath correctly mela khaata hai. Ecocline paaristhitik system (ecosystems) se related hai, Thermocline depression ke saath water ke Temperature me fastee se badalaav se related hai; Halocline depression ke saath lavanata (salinity) me fastee se badalaav se related hai. Pycnocline wo layer hai jo water nikaay me ghanatv pravanata (density gradient) ko darshaatee hai. Halocline lavanata pravanata ko show karta hai jabki Thermocline Temperature sankraman layer ko represent karta hai. *[Q54–Q55]*
- "main Skygangaa (milky way) ki naagarik hoon", is deeper statement ka shreya astronaut aur Space Shuttle mission visheshajny Kalpana Chawala ko diya gaya hai, jinakee kolanbiy aapada me mrityu ho gayi thi. India government ke nuclear energy department (DAE) ne 1962 me neshanal phartilaaijars limited ke parisar me panjaab ke Nangal me pehla heavy water sandevice chaaloo kiya tha. *[Q56–Q57]*

⚠ EXAM TRAP — Q46–Q47 — Bharati Research Station

▸ India ki third permanent Antarctic research station ka standard spelling “Bharati” hai; older question papers me “Bharti” variant mil sakta hai.

5. Magnetism, Timekeeping, MOEMS, Radar & Drones

- Anumagnettv (Paramagnetism) magnettv ka ek roop hai jisse kuch material baahya roop se laagoo magnetic field dwara lessjor roop se aakarshit hoti hain. Oxygen apne ayugmit (unpaired) electrons ki wajah se anumagnetic matter ka ek utkrisht example hai. Iron lauha-magnetic (ferromagnetic) hai, jabki Hydrogen aur Nitrogen prati-magnetic (diamagnetic) hain. Iron, Nickel aur Cobalt lauha-magnetic (ferromagnetic) tatva hain aur magnet ki direction majabootee se aakarshit hote hain. Aluminium ek anumagnetic tatva hai, jiska meaning hai ki ye anivaary roop se gaira-magnetic hai aur general magnet se aasaanee se aakarshit nahi hota hai. *[Q58–Q59]*
- options me se, Copper prati-magnetic (gaira-magnetic) hai, jiska meaning hai ki ye magnet ki direction aakarshit nahi hota hai. Nickel aur Cobalt lauha-magnetic hain. Lesspaas ki magnetic suee Earth ke magnetic field ke saath khuda ko sanrekhit karti hai; aama taur par magnetic Northee dhruva (North dishaa) ki direction ishaara karti hai. *[Q60–Q61]*
- Tepa rikordar ke tepa me ek plaastik baiking hoti hai jise pheromaignetik paaudar (jaise aayaran oksaaida) ke chhote particles ki ek patalee layer se lepita kiya jata hai. Rikording ke time, ye particle audio signal ko stora karne ke liye magnettv mila karte hain. Scientists (jaise JILA) dwara developed world ki sabse accurate nuclear ghadiyan abhootaEast accurateta ke saath time rakhne ke liye ek optical jaalee (lattice) me Strontium atomson ka use karti hain. *[Q62–Q63]*
- Automatic clocks (selpha-vaaindinga ghadiyan) ko Battery ki requirement nahi hoti hai. Ve main spring (mainspring) ko ghumaane ke liye pahanane wale ki kalaaee aur baanha ki praakritik speed se speedj energy (kinetic energy) ka use karte hain, jo ghadee ko power provide karta hai. ek mechanical (mechanical) ghadee me, pendulam jhoolate hue time ko control karta hai. ek modern electronic (kvaartja) ghadee me, ek 'Crystal oscillator' voltej laagoo hone par ek accurate, sthira frequency (frequency) par kanpan karke time measure karne wale tatva ki tarah kaam karta hai. *[Q64–Q65]*

- Kvaartj ghadiyan kvaartj kristal ke ek chhote tukade ka use karti hain jo electrical flow flowit hone par ek accurate frequency par kanpan karti hai. Ye Piezoelectric effect (mechanical tanaav ke javaab me electrical aavesh utpann karne ya electrical field ke javaab me kanpan karne ki capacity) par depend karta hai. MOEMS ka full form 'Micro-Opto-Electro-Mechanical-Systems' hai. optical emaeeemaes ke roop me bhi jana jata hai, ve chhote device hain jo prakaashikee (optics), electronics aur mechanicalee ko milaate hain. *[Q66–Q67]*
- Robotics ke context me, PUMA ka meaning 'programebala yoonivarsal machine phora asenbalee' (Programmable Universal Machine for Assembly) hai. Ye ek famous industrial robot aarma hai. Weather radar aasaanee se baarish ya barpha jaisee varsha ki nigaraanee karte hain (speedvidhi 2). Radar flight bharane wale jaanavaron jaise pakshiyon aur chamagaadadon ki speed ka detect lagaakar unke pravaas ko bhi traika kar sakte hain (speedvidhi 3). *[Q68–Q69]*
- Bade phasal kshetron me accurate roop se keetanaashakon ka chhidakaav karne ke liye Agriculture drone ka use kiya jata hai. (1) Kaimare aur sensar se equipped drone ka use niyamit roop se active jvaalaamukhiyon ke kretar jaise hazardous kshetron ka safe roop se nireekshan karne ke liye kiya jata hai. (2) *[Q70]*

⚠ EXAM TRAP — Q59 — Aluminium and Magnetism

▸ Aluminium technically paramagnetic hai, lekin attraction bahut weak hota hai; ordinary magnet ke saath practically “not noticeably attracted” ke roop me exam framing aa sakti hai.

6. Research Institutes, Technology Centres & General Science Applications

- Central electrical chemical research institute (CECRI) 14 janavaree, 1953 ko ek physical vaastavikata bana gaya, jab do esa raadhaakrishnan dwara karaaeekudee (Tamil Nadu) me ise raashtr ko dedicated kiya gaya. Bengaluru me Bhartiya science institute (IISc) ki establishment varsha 1909 me jamashedajee Tata ke active sammeaningn aur dooradarshee prayason se ki gayi thi. *[Q71–Q72]*
- RKCL - Rajasthan Knowledge Corporation Limited raajasthan government dwara established ek pablik limited kanpanee hai. Kanpanee ka main purpose ek new shaikshik dhaancha developed karna hai jo raajasthan raajya me 21veen century me aaeetee kaushal ki developmentasheel aavashyakataaon ki yojanaa, kaaryanvavehicle, paryavekshan aur niyaman kar sake. National Earth science study hub (NCESS) Earth science me scientist aur technical research aur development study ko badhaava dene ke liye ek svaayatt research hub hai. Ise Kerala government dwara 1978 me tiruvanantapurama, Kerala me CESS ke roop me establish kiya gaya tha. *[Q73–Q74]*
- Neshanal sentar phora antaarkatik enda oshana Research (NCAOR) ek Bhartiya research aur development institute hai, jo Bengaluru me nahi balki vaasko dee gaamaa, govaa me located hai. Baakee correctly matched hain. Central Bhartiya language institute ek Bhartiya research aur shikshan institute hai jo Mumbai me nahi balki maisoor (Karnataka) me located hai. Other die gaye jode correctly matched hain. *[Q75–Q76]*
- Central high tibbatee study institute - vaaraanasee Indira gaandhee development research institute - Mumbai National praakritik chikitsa institute (NIN) Pune, Maharashtra me located hai. (nota: die gaye source paatha me iska North Bangalore bataay gaya tha, jo historically kisi specific raajya-stareeya institute ko refer kar sakta hai, lekin major National institute Pune me hai). *[Q77–Q78]*
- Haaee-teka sitee - Hyderabad Saains sitee - kalakatta (Kolkata) Maaikrophon ek Soundka (acoustic) se electric traansadyoosar ya sensar hai jo Sound ko electrical signal (electrical signal) me convert karta hai. Early Telephone aur Telephone ripeetars me kaarban maaikrophon ka widely use kiya jata tha. *[Q79–Q80]*
- Planeton ki speed ke teen niyama 1600 ke decade ki shuruaat me johaans Kepler dwara prastaavit kiye gaye the. Kepler ne ye clearly kiya ki pratyek planet ek andaakaar orbit me Sun ke charon direction orbit karta hai. Solekasha (Soleckshaw) ek paryavarana-anukoola Solar rickshaw hai. Ye aanshik roop se paidal dwara aur aanshik roop se Battery dwara aapoorti ki jane wali electrical power dwara operated hota hai jo saura energy se chaarj hota hai. Ise CSIR laboratory me scientists ki ek teema dwara develop kiya gaya tha. *[Q81–Q82]*
- Bibliyometree (Bibliometry) ek Information Management Tool hai. Ye jnyaan ke production aur prasaar ke level ko measure karne ke saath-saath ek scientist field ke development ko traika karne ke liye ek maatraatmak saankhyikeey technology hai. Saaitotron wo device hai jisadwar Artificial climate utpann ki jati hai. Nota: saaitotron ek chikitsa device ka bhi naam hai jise ootaka engineering aur kainsar upachaar ke liye RFQMR use karke India me develop kiya gaya hai. *[Q83–Q84]*
- Roorkee koleja (Roorkee College) ki establishment 1847 ee. Me British saamraajy ke pehle engineering koleja ke roop me hui thi. Ise aba aaeeaaeetee Roorkee ke naam se jana jata hai, jo Uttarakhand me located hai. *[Q85]*

⚠ EXAM TRAP — Q78 & Q84 — Source/Name Ambiguity

▸ Q78 ke source note ke mutabik National Institute of Naturopathy (NIN) Pune me hai; older/official-key wording historically vary kar sakti hai.

▸ “Cytotron” term alag contexts me use hua hai; source question artificial-climate device ko target karta hai, jabki medical-device trade-name context bhi milta hai.

7. Metric System, Scientists, Laser Applications, Cryonics & Nuclear Basics

- India me meetrik system (dashamalava system) 1 aprail 1957 ko start ki gayi thi. 1957 se 1964 ke beech dhaale gaye sikkon ko 'new paisaa' kahaa jata tha. Taichyon (Tachyon) ek greeka word hai jiska meaning teevra hota hai. Physics me, taichyon ek kaalpanik particle hai jo hamesha light ki speed se zyada fastee se yatraa karta hai. *[Q86–Q87]*
- Johannes see. Maathara: physics me 2006 ke Nobel Prize vijeta (A-4). Maaikal griphina: NASA ke prashaasak (2005-2009) (B-3). Do. Raajaa ramanna ek prakhyat nuclear physicist the, jinhone 1974 me India ke pehle shaantipoorn nuclear prayog (smaailinga buddhaa) ke liye teema ka nirdeshan kiya tha. Unke baare me sabhi statement sahi hain. *[Q88–Q89]*

- Vaatar jeta teknolojee water ke bahut high Pressure wale jeta use karke vibhinn type ki saamagriyon ko kaatane me saksham hai. Is technology ka use mainly Drilling of mines (khudaaee) aur vaimaanikee me kiya jata hai. Lejara beema ka use kainsar (tyoomara vinaasha), eyes ki sarjaree (jaise LASIK); gurde ki patharee ko todne sahita kai beemaariyon ke upachaar me widely kiya jata hai. Isliye, ek se zyada option sahi hain. *[Q90–Q91]*
- Sookhee barpha - thosa kaarban daaioksaaid (A-3). Jeena therepee - rakta rogon/aanuvanshika vikaaron ka upachaar (B-4). Edoba (Adobe) systems ek American bahuNational computer software kanpanee hai jo multimedi aur software utpaadon (jaise photoshopa, peedeeeph reedara) ke development par kendrit hai, hardware par nahi. Isliye, option (D) correctly matched nahi hai. *[Q92–Q93]*
- Kaarban monooksaaid ek highly vishailee gas hai jo ganbhir air pradooshan paidaa karti hai. Ye statement true hai. Saimual haineemain homyopaithi ke sansthapak the. Chaarls daarvin ne developmentavaad ke principle ka pratipaadan kiya tha. Christopher kolanbas ne America ki discover ki thi. Isliye, kevala (C) true hai. *[Q94–Q95]*
- Jaisaa ki established hai, nuclear bama (atom bomb) Nuclear fission (heavy nuclear naabhik ke vibhaajana) ke principle par kaam karta hai. Hydrogen bama Nuclear fusion par kaam karte hain. 9 agasta 1945 ko naagaasaakee par giraae gaye Plutonium-implojana (Plutonium) type ke nuclear bama ka code naam 'phaita maina' (Fat Man) tha. Doosri direction, 6 agasta 1945 ko hiroshima par giraae gaye 'litila boya' me Uranium-235 ka istemaal kiya gaya tha. *[Q96–Q97]*

⚠ EXAM TRAP — Q91 & Q95 — Multiple/Loose Wording

▸ Laser beam medicine me multiple applications rakhta hai (eye surgery, kidney-stone treatment, tumour therapy etc.); Q91 source itself multiple-correct mark karta hai.

▸ Q95 ka wording loose hai: Otto Hahn aur Fritz Strassmann ne uranium nuclear fission experimentally establish ki; atomic bomb fission principle par based hai.

8. Displays, Automobiles, Humanoid Robots & Robotics

- 29 April, 1953 ko, losa enjils (America) ke television Station ne 3D me vijnyaana-katha programme 'space paitrola' ka ek episod transmit kiya. Ye world ka pehla 3D television prasaaran tha, jiske liye vishesh polaroid chashme ki requirement thi. Statement 1 and 2 sahi hain. Elaseedee (Liquid crystal disple) me Liquid crystal ka heavy use kiya jata hai. Internet eksesa ke liye modema computer ko phone/kebala laainon se connect karta hai. *[Q98–Q99]*
- Liquid crystal ka widely modern electronics me Liquid crystal disple (LCD) banane ke liye use kiya jata hai. Isme kalaaee clocks, teevee aur monitor ke disple skreena, Pocket Calculators aur portable computer me unakaa use shamil hai. Isliye sabhi option sahi hain. ek OLED me ek utsarjak ilektro-lyoominasenta layer hoti hai jo kaarbanik yaugik ki ek philma hoti hai jo light emit karti hai. Choonki OLED ko kathor baikalaait (LCD ke opposite) ki requirement nahi hoti hai, isliye unhe lacheele plaastik sabsatret par gadhaa ja sakta hai, kapadon ke liye rola-apa disple me banaye ja sakta hai; yahaan tak ki paaradarshee bhi banaye ja sakta hai. Isliye, sabhi statement sahi hain. *[Q100–Q101]*
- 1080p (jise phula echadee / Full HD bhi kaha jata hai) echadeeteevee haaee-dephinishana video moda ka ek seta hai. Standard vaaidaskreen 16:9 pahaloo anupaat ke saath, 1080p kshaitij roop se 1920 piksal aur lanbavat 1080 piksal (1920 x 1080) ke rijolyooshan me badala jata hai. Radiator (Radiators) hit eksachenjars hain jinka use mainly otomobile me internal dahana engine ko thanda karne ke liye kiya jata hai. Ve engine koolent se garmee ko external air me sthanaantarit karte hain, jisse engine ko zyada garma hone se rokaa ja sakta hai. *[Q102–Q103]*
- Mobile aur otomobile ne Bhartiyon ke social jeevan me ek badi kraanti laa dee hai. Mobile ne communication aur kanektivitee (2) me kaaphee improvements kiya hai, jabki otomobile ne logon ki shaareerik speedsheelata aur parivahan (1) ko kaaphee badhaay hai. Ye directly taur par "sanvedanasheelataa" se related nahi hai. 'Manav' (Manav) India ka pehla 3D-printeda hyoomanoid robot hai. Ise 2014 ke end me e-seta trening enda Research Institute ki laboratory me divaakar vaishy dwara develop kiya gaya tha. Ye 2 phuta lanbaa robot hai jisne Mumbai me IIT-bombe tekaphest me apni shuruaat ki thi. Isme drishti aur Sound prasanskaran capacityen hain. *[Q104–Q105]*
- Teslaa ke seeeeo elona maska ne aktoobar 2022 me teslaa ke eaaee de par apne hyoomanoid robot 'Optimus' (Optimus - jise teslaa bota ke roop me bhi jana jata hai) ke prototype ka khulaasa kiya tha. Optimus teslaa kaaron ke otopaayalat draaivar sahaayata suvidhaaon ke saath kuch eaaee software aur sensar share karta hai. Haalaanki robot ke kai laabha hain (vikalaanga vyaktiyon ki sahaayata karna, hazardous paristhitiyon me kaam karna, athaka roop se kaam karna), lekin ye tathya ki "ve naukariyon ko pratisthapit (replace) kar sakte hain" ek saamaajika-aarthika damage hai, kyonki isase Manav shramikon ki naukariyan chhina jati hain. *[Q106–Q107]*
- All India Council for Robotics and Automation (AICRA) ek gaira-laabhakaaree organisation hai jo India me robotics aur otomeshan me standard established karne ke liye sheersh nikaay ki tarah kaam karta hai. Iski establishment 2014 me hui thi. *[Q108]*

✓ Source coverage: Q1–Q108